
1. Relational Databases (RDBMS)

- **Structure:** Tables (rows & columns) with predefined schema
- **Use case:** Transactional systems, structured data

Examples:

- MySQL
- PostgreSQL
- Oracle
- SQL Server
- Amazon RDS / Aurora

 **Use When:** You need ACID transactions, SQL queries, strong consistency

2. NoSQL Databases

Designed for unstructured or semi-structured data, high scalability

♦ a) Document Databases

- Data stored as JSON-like documents
- Flexible schema

Examples:

- MongoDB
- Amazon DocumentDB

- CouchDB

 **Use When:** Storing user profiles, catalogs, CMS

♦ b) Key-Value Stores

- Simple key-value pairs (like a dictionary)
- Very fast for lookups

Examples:

- Redis
- Amazon DynamoDB
- Riak

 **Use When:** Caching, session storage, real-time leaderboards

♦ c) Columnar Databases

- Store data by column rather than row
- Optimized for analytics and OLAP

Examples:

- Apache Cassandra
- Amazon Redshift
- HBase

 **Use When:** Big data analytics, log analysis, time-series data

♦ d) Graph Databases

- Store entities and relationships as nodes and edges

Examples:

- Neo4j
- Amazon Neptune
- ArangoDB

 **Use When:** Social networks, recommendation engines, fraud detection



3. Time-Series Databases

- Optimized for timestamped data (metrics, logs, events)

Examples:

- InfluxDB
- Amazon Timestream
- Prometheus

 **Use When:** Monitoring systems, IoT sensor data, financial data



4. Object-Oriented Databases

- Data is stored as objects (used in OOP)
- Less common, tightly coupled with applications

Examples:

- db4o
 - ObjectDB
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5. In-Memory Databases

- Keep data in RAM for ultra-fast access

Examples:

- Redis
- Memcached
- VoltDB

 **Use When:** Real-time analytics, caching, gaming

6. Distributed / Cloud Databases

- Designed to scale across multiple servers/cloud regions

Examples:

- Amazon Aurora (distributed RDBMS)
- Google Spanner
- CockroachDB
- YugabyteDB

 **Use When:** High availability, global distribution
